

Preliminary sizing and optimisation of a satellite thermal control system using THERMICA

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1 Introduction

The thermal control system (TCS) is a critical subsystem in spacecraft engineering, ensuring that all components operate within their allowable temperature ranges despite the harsh environment of space. In this project, the preliminary sizing and optimization of a TCS for a cube-shaped satellite are performed. The primary objective is to maintain the temperature of onboard equipment between $-10\text{ }^{\circ}\text{C}$ and $+40\text{ }^{\circ}\text{C}$ across four extreme thermal scenarios: Beginning-of-Life (BOL) and End-of-Life (EOL) during both Winter and Summer Solstices.

2 Methodology

The analysis is divided into two parts. First, a preliminary analytical design is conducted to size radiators and heaters based on steady-state energy balance calculations. Second, these design parameters are implemented and verified using the Systema/Thermica software suite. This numerical analysis includes orbital simulations to evaluate the impact of internal radiation, optimize heater duty cycles, and verify compliance in survival modes.

3 First Part

There exists a total of 4 separate extreme thermal cases that the system will experience, these are

- Beginning-of-Life (BOL) - Winter solstice
- Beginning-of-Life (BOL) - Summer solstice
- End-of-Life (EOL) - Winter solstice
- End-of-Life (EOL) - Summer solstice

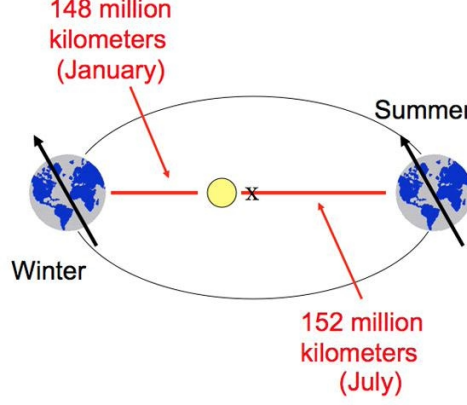


Figure 1: Position of the Earth w.r.t. the Sun during Summer and Winter

Because the absorptivity α is lower in the BOL configuration, the system absorbs less incident solar radiation. In addition, the Earth–Sun distance is greatest at the summer solstice, further reducing the available solar flux. Consequently, the BOL–Summer Solstice condition represents the coldest-case scenario for the system.

At end of life, when α reaches its maximum value and the Earth–Sun distance is at its minimum during the winter solstice, the system experiences the highest incident thermal load. Therefore, the EOL–Winter Solstice condition corresponds to the hottest-case scenario.

To simulate the effects of a satellite orbiting the Earth, a 6-sided cube was created in Thermica, with each side consisting of 1 m² radiators of the material SSM Aluminum, with emissivity coefficient of $\epsilon = 0.78$ and absorptivity of $\alpha = 0.15$ at BOL and $\alpha = 0.19$ at EOL due to material aging. The satellite was tested at a sun-synchronous orbit around the Earth at an altitude of 830 km with a mean solar time at the ascending node of 22h30, for two different attitude modes:

- *Nominal*: -Z axis towards Earth, -Y in the direction of satellite velocity,
- *Survival*: -Z axis towards the Sun, TTC antenna pointing to the Earth, satellite spinning around +Z.

Following the instructions provided, the mean thermal fluxes from the Sun, the Earth's albedo and the Earth's Infrared Radiation (IR) were calculated for the extreme cases previously mentioned and averaged over an orbit, and the results are presented in tables 1 and 2 for the sides with the different orientations $\pm X$, Y and Z of the cube satellite.

Source	-Y	X	Y	-X	Z	-Z
Sun	45.13	38.81	45.16	0	60.61	8.07
Earth albedo	4.53	4.80	4.53	4.21	0	16.51
Earth IR	35.96	35.96	35.96	35.96	0	131.14

Table 1: Incoming thermal energy flux during BOL summer solstice in W/m².

Source	-Y	X	Y	-X	Z	-Z
Sun	59.48	72.62	59.48	0	78.80	11.78
Earth albedo	5.87	6.39	5.87	5.29	0	21.41
Earth IR	35.96	35.96	35.96	35.96	0	131.14

Table 2: Incoming thermal energy flux during EOL winter solstice in W/m².

And as expected the BOL summer solstice case provides less heat from the surrounding environment than in the EOL winter solstice case.

To ensure that electronics and vital equipment remain at an operational temperature within the requirements, the walls of each side should not exceed 20 °C and not go below 0 °C, which leaves a margin for the actual limits of -10 °C to 40 °C. To better understand how the heat was distributed at each wall and to optimize the placement of electronic equipment, the thermal rejection capacity, i.e. the emitted minus the absorbed thermal flux, was calculated for each of the walls in the coldest and hottest cases. By assuming that the surrounding environment acts as a cold sink at 0 K, and that each wall radiates according to a gray body, the emitted heat from each side follows Stefan-Boltzmann's law, and can be calculated using

$$E_e = \sigma \epsilon T^4, \quad (1)$$

where $\sigma = 5.670 \cdot 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$, ϵ is the emissivity coefficient, and T is the temperature of the side. The results of the rejection capacities are presented in table 3.

Case	-Y	X	Y	-X	Z	-Z
Hottest	224.64	210.98	224.64	284.70	247.15	161.62
Coldest	160.04	166.09	160.01	205.49	185.05	89.94

Table 3: Heat rejection capacities of the different walls of the cube in W/m².

Identifying which external surfaces have higher heat-rejection capability is essential for optimal equipment placement, particularly for components with significant power dissipation. This ensures that thermal loads are allocated to surfaces best suited to reject the generated heat. The sides $\pm X$ are assumed to mount solar panels and instruments, and have MLI, which prevents any heat exchange with the surrounding environments. The equipment used in this simulation is listed in table 4.

Function	$T_{\min}$ (°C)	$T_{\max}$ (°C)	Mass (kg)	Nom. mode	Surv. mode	
Power supply	-10	40	40	150 / 50 W	20 W	+Z
On-b. data proc.	-10	40	5	15 / 10 W	10 W	-Z
AOCS	-10	40	30	110 / 50 W	30 W	+Y
Telemetry	-10	40	20	70 / 0 W	20 W	-Y

Table 4: Thermal, mass, and power characteristics of the main subsystems.

The equipment layout was defined based on the heat-rejection capabilities under the

hottest operating case and the corresponding power dissipation of each unit. Components with the highest power dissipation were assigned to the remaining surfaces (i.e., the $\pm Y$ and $\pm Z$ panels) with the greatest heat-rejection capacity, followed by lower-dissipating units on surfaces with progressively lower rejection capability.

Then, it is assumed that a fraction of each of the sides (excluding $\pm X$) are covered with radiators, and the rest is covered with MLI's. To find the fraction required for each side, a balance needs to be found at the hottest case where the emitted power is equal to the absorbed power (both from the environment and onboard electronics) when the temperature of each side is 20 °C. This can be modeled by the following that describe the net heat flux for each face:

$$E_{emitted}A_{rad} - E_{absorbed}A_{rad} - P_{electronics} = 0 \quad \Rightarrow \quad A_{rad} = \frac{P_{electronics}}{E_{emitted} - E_{absorbed}} \quad (2)$$

where $E_{emitted}$ is the emitted radiation flux ($\sigma\epsilon T_{max}^4$), and the denominator represents the rejection capacity for the hot case. This results that the required area fractions for radiators on each side is according to table 5.

	-Y	X	Y	-X	Z	-Z
Area (m ²)	0.3116	0	0.4897	0	0.6069	0.0928

Table 5: The area fraction of radiators on each side of the cube.

The radiators' sizing is followed by the assumption that a heater is placed on each of the four walls $\pm Y$ and $\pm Z$. To determine at what power these heaters need to be operating, the coldest case scenario is considered with the purpose of keeping the temperature above 0 °C for each side. A similar power balance equation is used, with the addition of the term P_{heat} to describe the amount of heat the heaters must provide. This yields the equation

$$P_{heat} = P_{emitted} - P_{absorbed} - P_{electronics} = A_{frac}(\sigma\epsilon T_{cold}^4 - E_{absorbed}) - P_{electronics} \quad (3)$$

and the results are shown in table table 6 where a negative value simply means that

Faces	X	-X	Y	-Y	Z	-Z
Power required (W)	0	0	28.3564	59.8379	62.3152	-1.6529

Table 6: Power required for the heaters on each side of the cube in W.

the analysed side absorbs more heat than it emits in the coldest case, not requiring any heating.

4 Second Part: Tasks

Introduction

For the second phase of the project, the radiator dimensions and heater power ratings calculated in Part 1 were implemented into the simulation software. These parameters were subsequently optimized to achieve the target heater duty cycles.

4.1 Thermal Network Diagram

Request: “Make a visual diagram of the nodal thermal network used by the thermo-mathematical solver to calculate the satellite temperatures. The diagram must show all thermal nodes and the thermal couplings (radiative and/or conductive) between nodes.”

The complete diagram of the nodal thermal network, illustrating all nodes and their respective radiative or conductive couplings, is shown in Figure A.1 in the Appendix.

4.2 Temperatures with NO internal radiation

Request: “Calculate the temperatures in the hottest case without internal radiation, with the radiators and heaters obtained during the preliminary design in part 1. For the conductive coupling between equipment and radiator, the value depends on the type of mechanical contact. You have 3 options and you shall choose the one you think is best:

- Thermal Joint: 40 W/K
- Equipment Screwed in Radiator: 10 W/K
- Equipment Insulated from Radiator: 0,6 W/K

”

For the conductive coupling has been choosen ”*Equipment Screwed in Radiator: 10 W/K*”, This value was selected to establish a trade-off between heat sinking and thermal isolation. It ensures sufficient heat transfer to dissipate power during hot cases, while preventing excessive coupling that could lead to under-temperature issues in cold scenarios. The resulting temperature diagrams for this configuration are presented in Figure A.2 in the Appendix.

General Results and Compliance

Based on the simulation results for the hottest case without internal radiation, the temperature evolution of the four equipment nodes was analyzed. The simulation shows that the maximum temperatures (T_{max}) for the equipment units are:

- Eq. Y^+ : **27.5 °C**
- Eq. Y^- : **20.5 °C**

- Eq. Z^+ : **30 °C**
- Eq. Z^- : **10.5 °C**

All temperatures are below the maximum qualification limit of **+40 °C** defined in the mission specifications. Therefore, the current thermal control system design is compliant with the absolute safety requirements.

Evaluating the thermal margins. The margin is defined as the difference between the qualification limit (40 °C) and the maximum calculated temperature.

- The critical path is represented by the equipment on the Z^+ face, which exhibits the smallest margin of 10 °C.

Impact of Internal Radiation and Coupling

It is important to note that this simulation does not include internal radiative exchange. As a result, each face of the satellite acts as an isolated system, unable to redistribute heat from high-dissipation units (like the Power Supply or AOCS) to cooler areas of the spacecraft structure. This represents a “worst-case” scenario for local hot spots.

Conclusions

In conclusion, while the design is safe, the lack of internal radiation leads to higher peak temperatures on specific nodes. We expect that introducing internal radiation in the next steps (Question 9) will homogenize the temperatures across the satellite, likely increasing the margins for the hottest units.

4.3 Temperature with internal radiation

Request: “Calculate the temperatures in the hot case with internal radiation. Compare with previous calculation and comment.”

The temperature graphs considering internal radiation are displayed in Figure A.3 in the Appendix.

Simulation Results

In this step, the internal radiation was activated to simulate the radiative exchange inside the satellite cavity. The simulation was performed for the hot case, and the resulting maximum temperatures (T_{max}) for the equipment are:

- Eq. Y^+ : **21.5 °C**
- Eq. Y^- : **18.5 °C**
- Eq. Z^+ : **21 °C**
- Eq. Z^- : **16.5 °C**

Comparison with Previous Calculation (Effect of Internal Radiation)

Comparing these results with those obtained in Question 8 (without internal radiation), we observe a general decrease in the peak temperatures of the hottest components. Specifically:

- The temperature of the previously Hottest Unit on the Z^+ face dropped by approximately $9 \Delta T$ °C.
- This effect is due to the “cavity effect”: the internal radiation allows heat to be redistributed from the hotter surfaces (equipment with high dissipation) to the cooler parts of the satellite (e.g., shadowed walls or lower dissipation areas).
- Consequently, the thermal gradients between the different faces of the satellite have been reduced, leading to a more homogenized thermal distribution.

Margins and Design Validation

With the inclusion of internal radiation, the margins against the qualification limit (40 °C) have improved.

- The minimum margin is now 18.5 °C on the equipment on the Y^+ face.
- Regarding the design target of 20 °C, the system is now closer to compliance compared to the simulation without internal radiation.

Conclusion

The activation of internal radiation demonstrates a beneficial effect on the satellite’s thermal control. By allowing radiative heat transfer between internal nodes, the system avoids excessive local overheating, bringing the temperature near the theoretical limits imposed by the preliminary analysis.

4.4 Cold case Temperatures with internal radiation

Request: “Calculate the temperatures in the cold case with internal radiation. You can plot heater powers by selecting ‘Residual flux’ in the 2D Curve view.9., Int.Rad. ON”

The corresponding temperature profiles and residual flux plots for the cold case analysis are provided in Figure A.4 in the Appendix.

Minimum Temperatures and Safety Limits

The simulation was performed for the cold case with internal radiation active. The primary goal was to verify that the equipment temperatures do not drop below the minimum qualification limit of **-10 °C**. The results show the following minimum temperatures (T_{min}):

- Eq. Y^+ : **-10 °C**
- Eq. Y^- : **-10 °C**

- Eq. Z^+ : **-10 °C**
- Eq. Z^- : **-9.25 °C**

Heater Activity Analysis (Residual Flux)

By analyzing the “Residual Flux” curves, which correspond to the power injected by the heaters, we observed the behavior of the active thermal control system:

- **Activation:** The heaters did activate during the coldest parts of the orbit.
- **Cycling:** The trend of the heaters can be observed in the graphs being a periodical switch ON and OFF to maintain the temperatures ≥ -10 °C.
- **Peak Power:** The maximum power consumption observed for the heaters was approximately 46 W.

Discussion on Control Logic

According to the script provided , the control logic is set to turn the heaters ON at **-10 °C** and OFF at **-5 °C**.

- The simulation confirms that the system **successfully maintained** the equipment with a $T \geq -10$ °C.

Conclusion

In the cold case, the thermal control system is compliant. The heaters are able to compensate for the heat loss during eclipses. However the margins are tight, considering that the equipment reached -10 °C, which is exactly on the qualification limit.

4.5 Duty Cycles optimization

Request: “Every time one heater is turned ON then OFF is called a heater cycle. During one period, the fraction of time with heater ON is called ‘duty cycle’. If the duty cycle is close to 100%, which is called saturation, it means there is no margin in case more heating power is needed. If the heating power is too low, it means the heating power is oversized. Optimise your TCS and try to get a maximum duty cycle between 60 and 70% for each heater.”

Iterative Optimization Process

To achieve the target duty cycles (between 60% and 70%), an iterative process was performed by changing the power of the heaters starting from the initial values obtained in the previous analysis. Specifically, the power was increased to lower the duty cycle, while it was decreased to increase the duty cycle. The table below summarizes the progression of Heater Power (P) and the resulting Duty Cycle (DC) for each face across the 3 iterations performed.

Step	Face Y^+ (Node 3)		Face Y^- (Node 4)		Face Z^+ (Node 5)		Face Z^- (Node 6)	
	P [W]	DC [%]	P [W]	DC [%]	P [W]	DC [%]	P [W]	DC [%]
Initial	28.36	46.7	59.84	82.0	62.32	68.5	0	-
Iter 1	20.36	74.0	64.84	75.5	62.32	68.5	0	-
Iter 2	22.36	58.8	68.84	67.5	62.32	68.5	0	-
Iter 3	21.36	61.9	68.84	67.5	62.32	68.5	0	-

Table 7: Evolution of Heater Power (P) and Duty Cycle (DC) throughout the optimization iterations.

Analysis of Results

As shown in the table:

- We started with an initial configuration where the duty cycles were too low or too high.
- By tuning the power on the faces with not on target duty cycles, we progressively converged towards the target range.

The final configuration (Iter 3) provides a robust margin for the cold case, ensuring that the heaters are neither permanently ON (saturation) nor significantly oversized.

4.6 Survival Mode Duty Cycles optimization

Request: “BONUS QUESTION: Now that the design has been optimized on the hot and cold cases, the survival mode can also be considered for cold case simulations. You have the choice to position the TTC antenna either on the $+y$ or $-y$ side; this antenna is to be pointed to the Earth as a secondary pointing constraint (primary pointing to the Sun, see specifications on survival attitude) while the SPOT satellite is in safe mode. After correctly setting the files for the simulation, validate your TCS design on safe mode. Can you get all the heaters to work with non-saturated duty cycles? What can be changed?”

The graphs of the residual fluxes for the survival mode are illustrated in Figure A.5 in the Appendix.

Optimization Data Table

The following table reports the exact values recorded during the 5 optimization iterations performed in the lab. The same optimization criteria as the previous task were used.

- **Iterations 1-4:** Focused primarily on fixing the saturation issue on Face Y^+ (Node 3).
- **Iteration 5:** Final adjustment of Node 3 and optimization of Node 4 and Node 5 to reach the target duty cycles.

Step	Face Y^+ (Node 3)		Face Y^- (Node 4)		Face Z^+ (Node 5)		Face Z^- (Node 6)	
	P [W]	DC [%]	P [W]	DC [%]	P [W]	DC [%]	P [W]	DC [%]
Start	21.35	Sat	68.83	19%	62.31	37%	0	-
Iter 1	30.35	Sat	65.83	20%	62.31	37%	0	-
Iter 2	60.35	Sat	60.83	21%	62.31	37%	0	-
Iter 3	70.35	80%	55.83	27%	62.31	37%	0	-
Iter 4	82.35	58%	35.83	52%	55.31	55%	0	-
Iter 5	80.35	61%	33.83	67%	50.31	62%	0	-

Table 8: Detailed log of the 5 optimization iterations (Data extracted from Lab notes).

5 Conclusion

The objective of this project was to design and optimize a Thermal Control System (TCS) for a cube satellite, ensuring equipment temperatures remain within the qualification limits of $-10\text{ }^{\circ}\text{C}$ to $+40\text{ }^{\circ}\text{C}$ across extreme orbital scenarios. Through preliminary analytical sizing and subsequent verification using Systema/Thermica, the design was shown to be compliant with all thermal requirements.

5.1 Summary of Results

The analytical sizing of radiators provided a strong baseline, which was refined during the simulation phase. In the **Hot Case** (EOL Winter Solstice), the initial simulation without internal radiation showed compliance, but with a relatively tight thermal margin of $10\text{ }^{\circ}\text{C}$ on the Z^+ face. The activation of internal radiation significantly improved the system’s performance. The “cavity effect” redistributed heat from high-dissipation units to cooler surfaces, reducing the peak temperature on the hottest node (Z^+) by approximately $9\text{ }^{\circ}\text{C}$. This confirmed that internal radiative exchange is critical for homogenizing thermal gradients and increasing safety margins.

In the **Cold Case** (BOL Summer Solstice), the heater sizing was validated. The active thermal control successfully maintained all nodes above the minimum limit of $-10\text{ }^{\circ}\text{C}$. Through an iterative optimization process, the heater duty cycles were adjusted from initial saturation (100%) or low efficiency to a target range of 60–70%. This optimization ensures that the satellite power system is not overburdened while retaining sufficient margin to handle colder-than-predicted anomalies.

5.2 Uncertainties and Validity of Assumptions

The credibility of these results relies on several key modeling assumptions:

- **Conductive Couplings:** A uniform conductive coupling of 10 W/K (representing equipment screwed into the radiator) was applied. This is a major source of uncertainty. If the mechanical interface has higher thermal resistance (e.g., due to roughness or insufficient torque), the equipment temperatures in the hot case would be higher than predicted, potentially violating the $+40\text{ }^{\circ}\text{C}$ limit.

- **Optical Properties:** The model assumes constant optical properties (α and ϵ) for BOL and EOL scenarios. In reality, surface degradation can be non-uniform, potentially leading to localized heating.
- **Geometric Simplification:** The 6-node thermal network is a simplified representation. It does not account for specific component placement *within* a face or the shading effects of external appendages (like deployed solar panels) on the radiators.

5.3 Future Improvements

To increase the fidelity of the thermal model and improve the design, the following steps are suggested for future work:

- Question 1: Transient Analysis:** The current analysis focuses heavily on steady-state or mean orbital behaviors. A transient analysis is needed to evaluate the rate of temperature change during eclipse transitions, which ensures that thermal shock does not damage the components.
- Question 2: Refined Heater Control:** The current simple On/Off logic resulted in temperatures hitting the -10°C limit exactly. Implementing a Proportional-Integral (PI) controller or adjusting the set points (e.g., turning on at -8°C) would provide a safer buffer against the hard qualification limit.

Appendix: Visual Diagrams and Plots

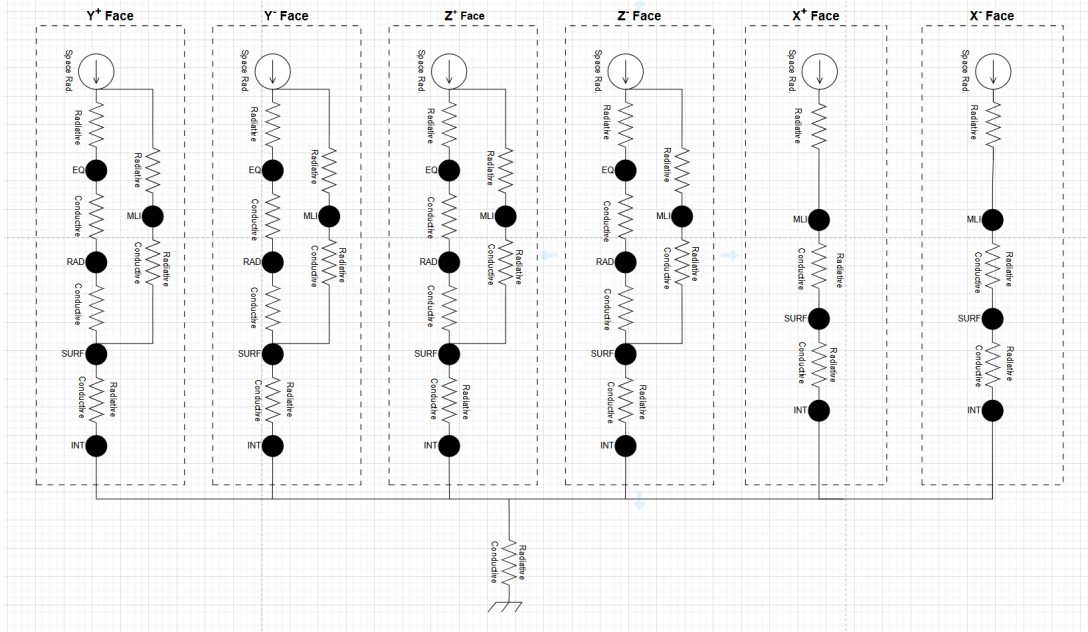
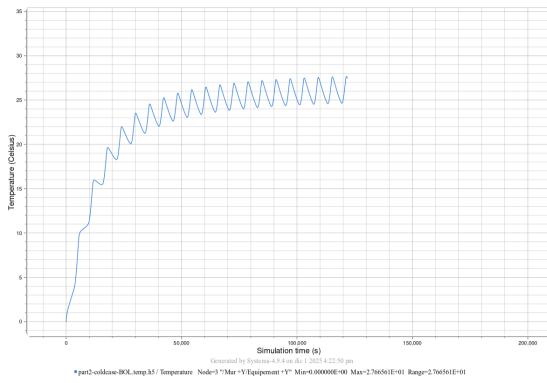
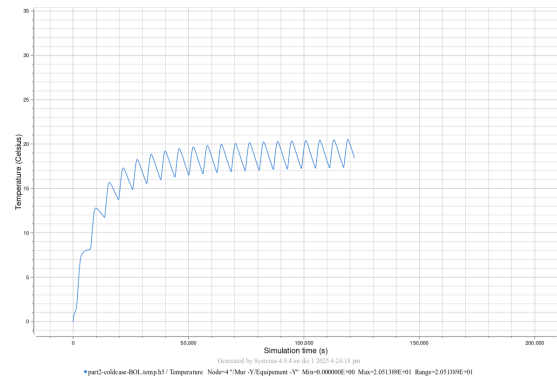


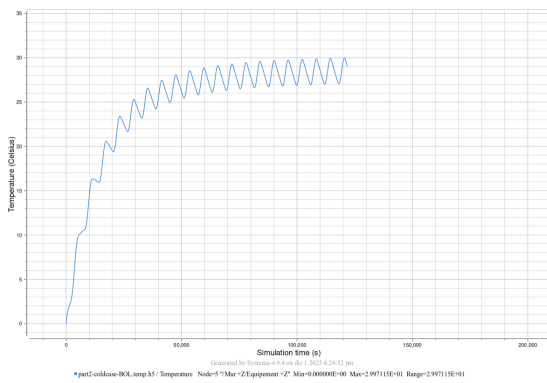
Figure A.1: Thermal Network of the satellite



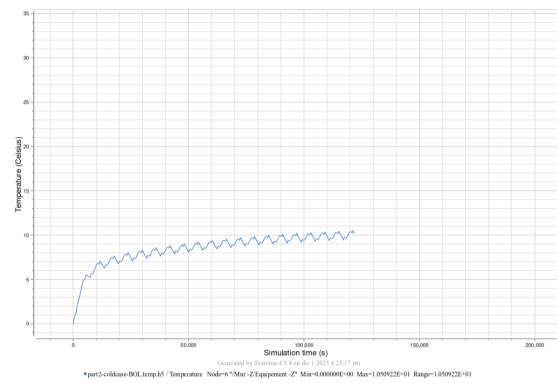
(a) Equipment on Y^+ face



(b) Equipment on Y^- face

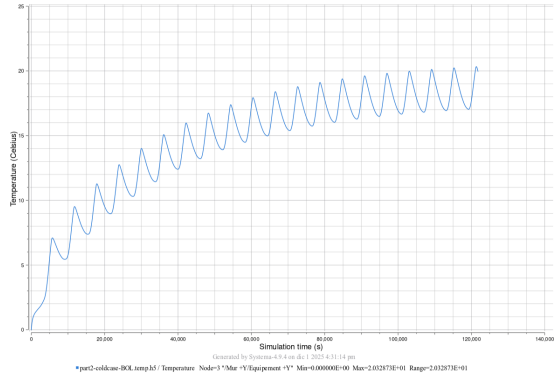


(c) Equipment on Z^+ face

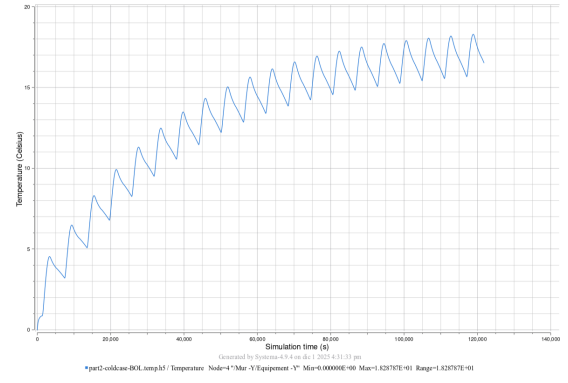


(d) Equipment on Z^- face

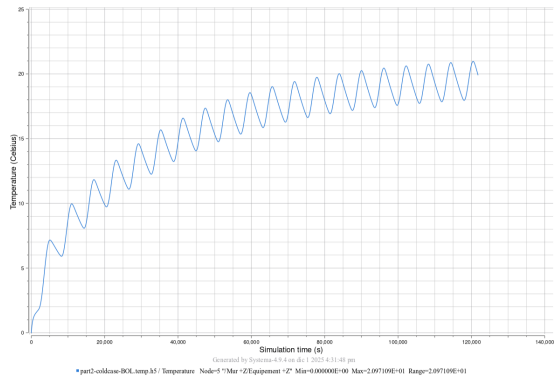
Figure A.2: Q.8 Hottest Case (No Internal Radiation) Temperature Diagrams.



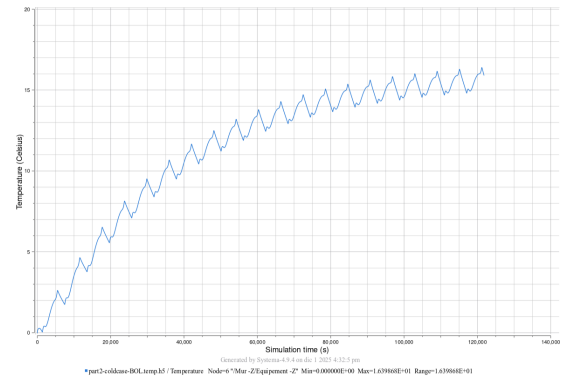
(a) Equipment on Y^+ face, Int.Rad. ON



(b) Equipment on Y^- face, Int.Rad. ON

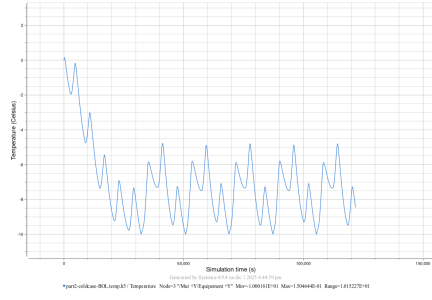


(c) Equipment on Z^+ face, Int.Rad. ON

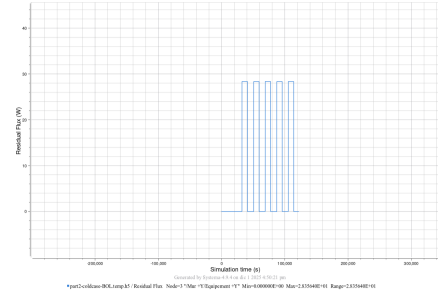


(d) Equipment on Z^- face, Int.Rad. ON

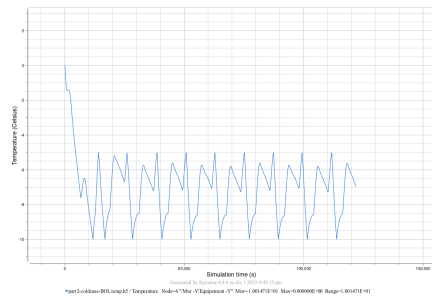
Figure A.3: Q.9 Hottest Case (WITH Internal Radiation) Temperature Diagrams.



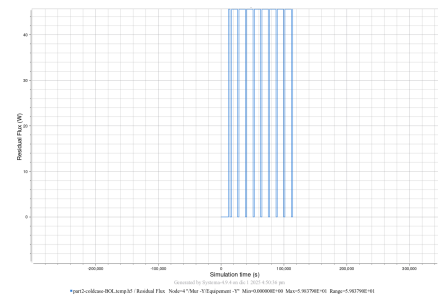
(a) Temp: Equipment on Face Y^+



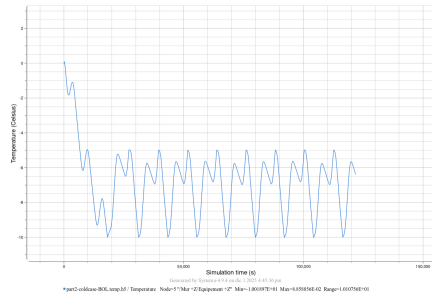
(b) Flux: Equipment on Face Y^+



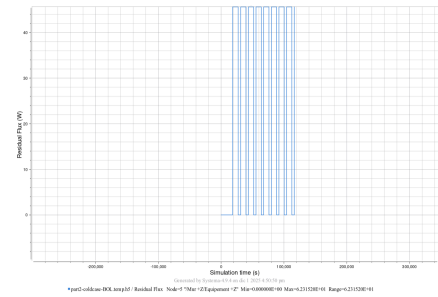
(c) Temp: Equipment on Face Y^-



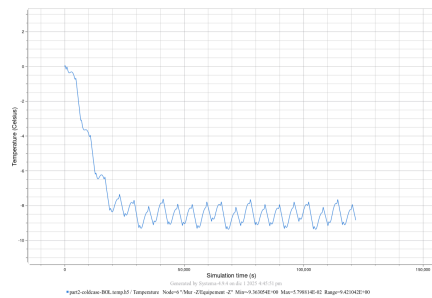
(d) Flux: Equipment on Face Y^-



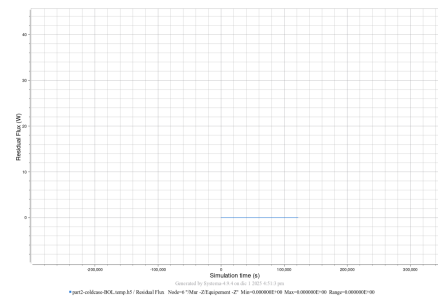
(e) Temp: Equipment on Face Z^+



(f) Flux: Equipment on Face Z^+

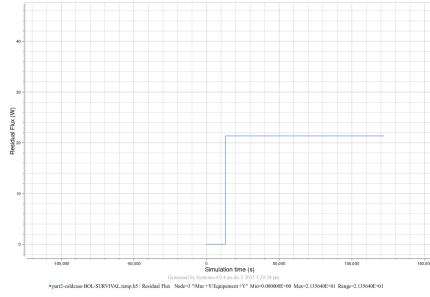


(g) Temp: Equipment on Face Z^-

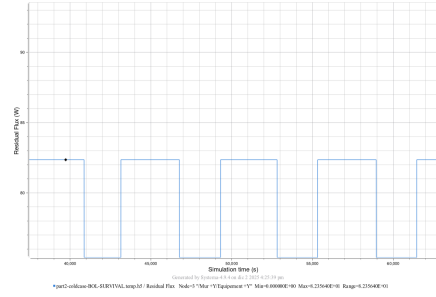


(h) Flux: Equipment on Face Z^-

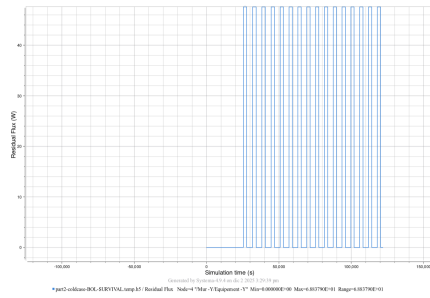
Figure A.4: Cold Case Analysis (Q.10): Temperature profiles (left column) and Heater Residual Flux (right column) for internal radiation scenario.



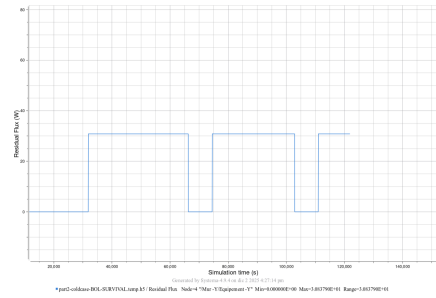
(a) Face Y^+ : Initial Flux



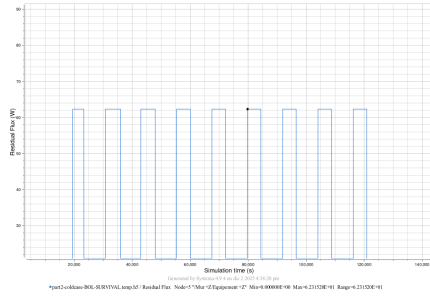
(b) Face Y^+ : Final Flux (Optimized)



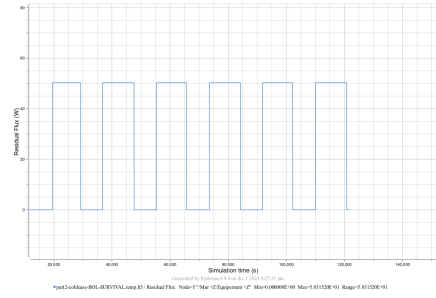
(c) Face Y^- : Initial Flux



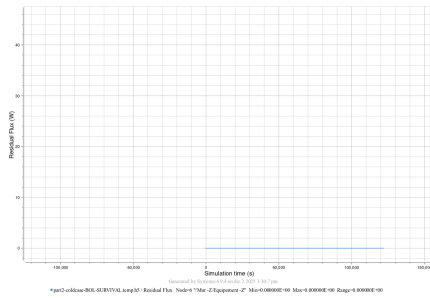
(d) Face Y^- : Final Flux (Optimized)



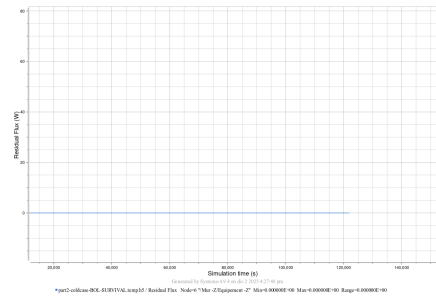
(e) Face Z^+ : Initial Flux



(f) Face Z^+ : Final Flux (Optimized)



(g) Face Z^- : Initial Flux



(h) Face Z^- : Final Flux (Optimized)

Figure A.5: Survival Mode Optimization: Comparison of Heater Residual Flux before optimization (Left) and after the final iteration (Right) for all equipment faces.

A Group Contributions

The contributions of each group member to the project work, presentation, peer review, and final report are summarized in Table 9.

Table 9: Contribution of each group member.

Name	Lab Work	Report	Presentation	Review
Brian Bertacco	<i>Yes</i>	<i>Part 2</i>	<i>Lab D</i>	<i>Yes</i>
Andreas Wangård	<i>Yes</i>	<i>Part 1</i>	<i>Lab C</i>	<i>Yes</i>
Jonas Planck	<i>Yes</i>	<i>Part 1</i>	<i>Lab C</i>	<i>Yes</i>
Samuele Cuzzi	<i>Yes</i>	<i>Part 1</i>	<i>Lab C</i>	<i>Yes</i>
Tobia De Benedictis	<i>Yes</i>	<i>Part 2</i>	<i>Lab D</i>	<i>Yes</i>
Felice Venuti	<i>Yes</i>	<i>Part 2</i>	<i>Lab D</i>	<i>Yes</i>

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